

Supplier Performance Evaluation Report

Client: VinSpire Beverages Pvt. Ltd.

Industry: Retail & Wholesale (Alcoholic Beverages)

About the Company:

VinSpire is a leading B2C distributor specializing in premium alcoholic beverages across the USA. Serving 80 retail outlets across major cities, VinSpire manages an extensive portfolio of wines, spirits, and liquors—including top-tier brands like Laphroaig, Cîroc, Glenfiddich, and Sauza. The company's mission is to maximize profitability through data-driven supplier management, inventory optimization, and strategic pricing in a highly regulated industry.

Business Problem:

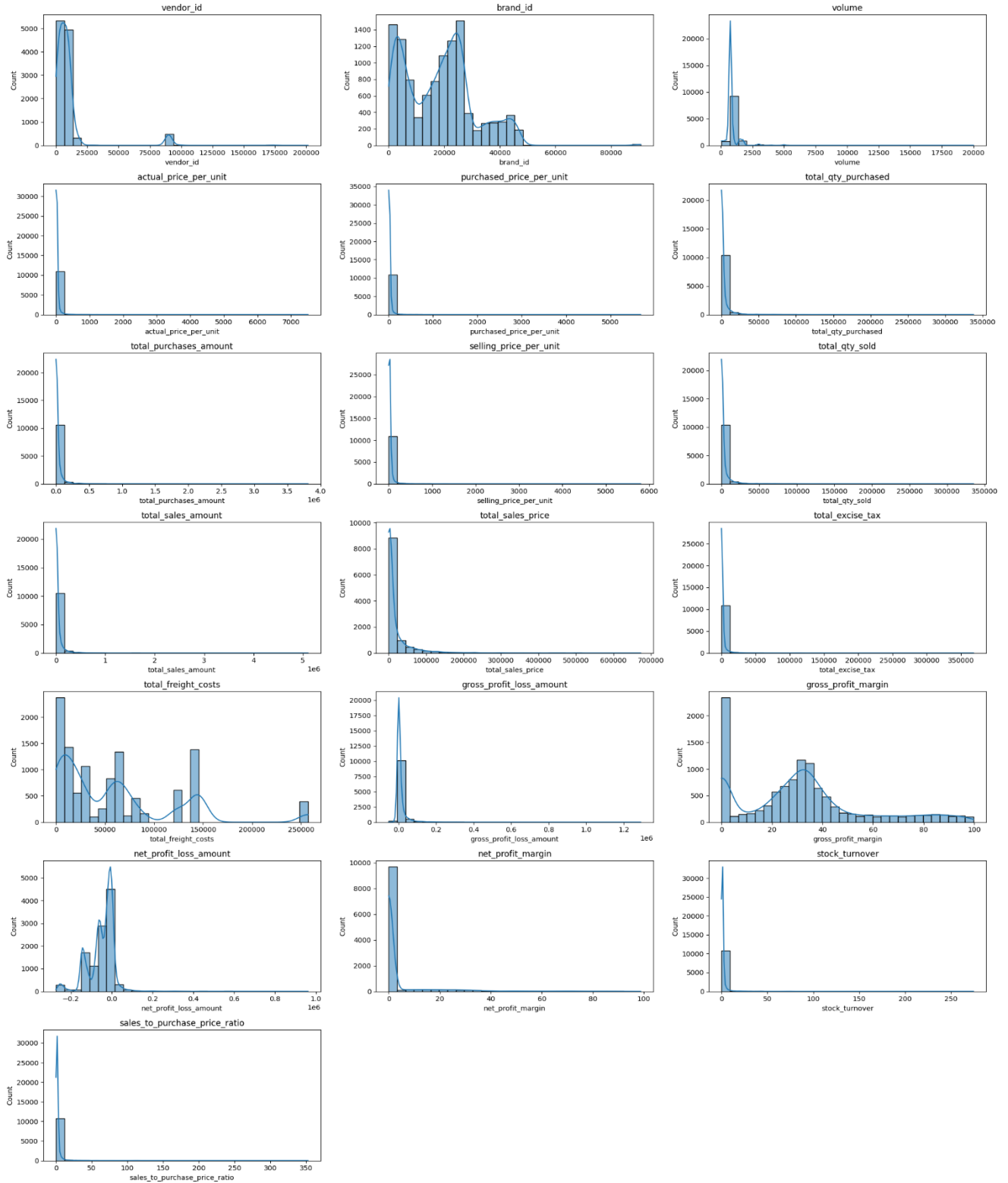
In the competitive alcoholic beverage distribution market, **effective inventory and sales management** are crucial for profitability. VinSpire relies on a diverse network of suppliers for its product offerings. Analyzing **supplier performance** allows the company to optimize inventory, improve cash flow, maximize gross profit, and strategically **align with high-performing supplier** critical in a highly regulated and competitive industry. The goal of this analysis is to identify the suppliers that help VinSpire **distinguish between low-performing and high-performing partners**.

EDA:

This focused on exploring clean and engineered data to uncover patterns, detect anomalies, and extract meaningful insights that align with the business objectives. Python's visualization libraries like **matplotlib and seaborn** and statistics were used to analyze distributions, relationships, and key metrics across suppliers and brands.

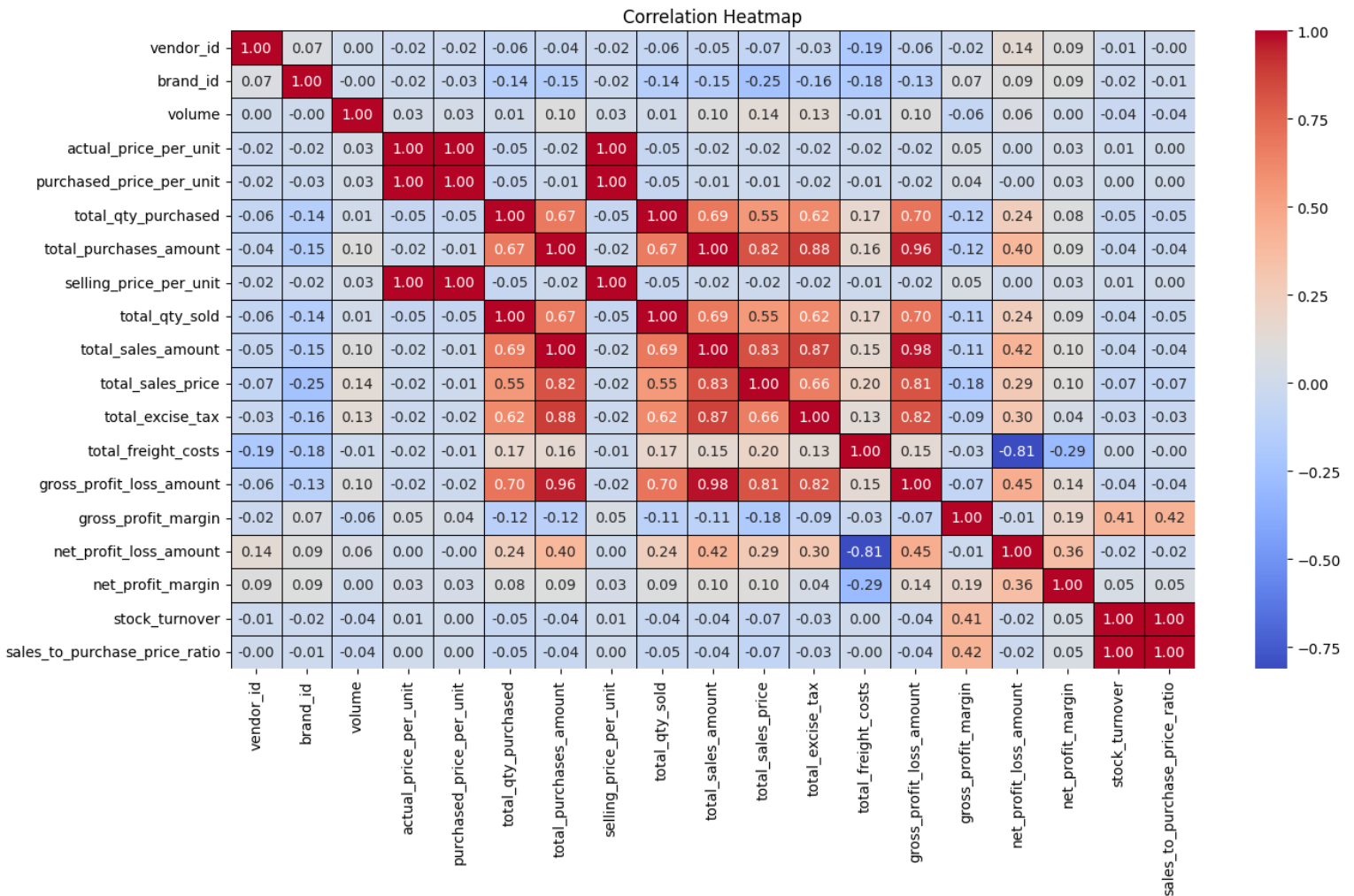
a) Summary Statistics:

	count	mean	std	min	25%	50%	75%	max
vendor_id	11101.0	10329.175930	18478.053301	2.00	3252.00	7153.00	9552.00	201359.00
brand_id	11101.0	18254.831006	12600.526806	58.00	6206.00	19070.00	25570.00	90631.00
volume	11101.0	844.887848	654.811062	50.00	750.00	750.00	750.00	20000.00
actual_price_per_unit	11101.0	36.629352	148.534699	0.49	10.99	15.99	28.99	7499.99
purchased_price_per_unit	11101.0	25.008177	109.078281	0.36	6.88	10.52	19.72	5681.81
total_qty_purchased	11101.0	3077.337807	10916.912273	1.00	34.00	250.00	1941.00	337660.00
total_purchases_amount	11101.0	29443.158015	120964.323734	0.71	446.16	3556.44	20025.94	3811251.60
selling_price_per_unit	11101.0	33.587459	126.334146	0.00	9.99	14.99	26.99	5799.99
total_qty_sold	11101.0	3015.154761	10777.015741	0.00	31.00	248.00	1879.00	334939.00
total_sales_amount	11101.0	41366.692946	164832.865883	0.00	725.67	5124.79	27534.38	5101919.51
total_sales_price	11101.0	18412.910834	44281.097727	0.00	289.72	2777.91	15665.36	672819.31
total_excise_tax	11101.0	1715.121743	10776.046210	0.00	4.59	44.10	396.60	368242.80
total_freight_costs	11101.0	59170.326662	60912.767016	0.09	11595.97	48347.26	79528.99	257032.07
gross_profit_loss_amount	11101.0	11923.534930	45490.515187	-52002.78	53.16	1389.35	8515.59	1290667.91
gross_profit_margin	11101.0	31.237934	24.744152	0.00	13.42	30.65	40.20	99.72
net_profit_loss_amount	11101.0	-48961.913475	66981.237999	-272403.27	-72377.34	-28812.17	-6089.99	961067.03
net_profit_margin	11101.0	3.358327	10.979329	0.00	0.00	0.00	0.00	98.76
stock_turnover	11101.0	1.722364	5.985775	0.00	0.81	0.98	1.04	274.50
sales_to_purchase_price_ratio	11101.0	2.530739	8.428293	0.00	1.16	1.44	1.67	352.93



- **Purchase & Actual Prices:** Maximum purchase (5,681.81) and sales prices (7,499.99) are far above the mean (24.39 and 35.64), suggesting the presence of premium or high-value products.
- **Total Quantity Purchased & Sold:** Extremely wide range (1 to 337,660 purchased; 0 to 334,939 sold) indicates both bulk transactions and very low-volume items.
- **Total Sales Quantity & Price:** Minimum values are 0, meaning some products were purchased but never sold. These could be slow-moving or obsolete stock.
- **Excise Tax:** Ranges from 0 to 368,242.80 — implying that tax is heavily volume-driven or varies by product category.
- **Freight Cost:** Highly variable (0.09 to 257,032.07), which may reflect logistics inefficiencies, inconsistent shipping methods, or batch-level freight entries.
- **Gross Profit:** Minimum value is -52,002.78, indicating losses. Some products or transactions may be selling at a loss due to high costs or selling at discounts lower than the purchase price.
- **Gross Profit Margin:** A minimum of 0% suggests some products generate no revenue, while 75% of vendors have $GPM \leq 40.20\%$, showing moderate profitability across most suppliers.
- **Net Profit Margin:** Much lower than GPM, with 75% of vendors having $NPM = 0\%$, showing that expenses like freight and excise tax significantly reduce net profits.
- **Stock Turnover:** Ranges from 0 to 274.5, imply some products sell extremely fast while others remain in stock indefinitely. Value more than 1 indicates that quantity sold for that product is higher than purchased quantity due to either sale being fulfilled from older stock.

b) Correlation Analysis:



- **Actual & Selling Price vs Total Sales:** The actual price and selling price have weak correlation with total sales (-0.02, -0.02). This shows price variations between the actual price and selling price do not impact on the total sales revenue.
- **Purchase vs Selling Prices:** Purchase Price has a strong correlation with Selling Price (+1) indicating that as the purchasing price increases, the selling price also increases.
- **Total Quantity Purchased vs Sold:** There is a very strong correlation between them (+1), confirming efficient inventory turnover.

- **Excise Tax:** Total excise tax has a strong correlation with total purchases (0.88), total sales (0.87) and gross profit/loss amount (0.82), indicating that as purchases, sales and gross profit increase, the overall excise tax to be paid also increases.
- **Freight Cost:** Total freight cost has a negative correlation with net profit/loss amount (-0.81), indicating that higher freight costs lead to a decrease in net profit.
- **Gross Profit Margin:** GPM has a slight negative correlation with total sales price (-0.11) indicating that GPM decreases slightly with an increase in total sales price. This could be due to competitive pricing in the market.
- **Stock Turnover:** Stock Turnover has a weak correlation with the net profit margin (0.05), indicating that faster turnover does not necessarily result in higher profitability.

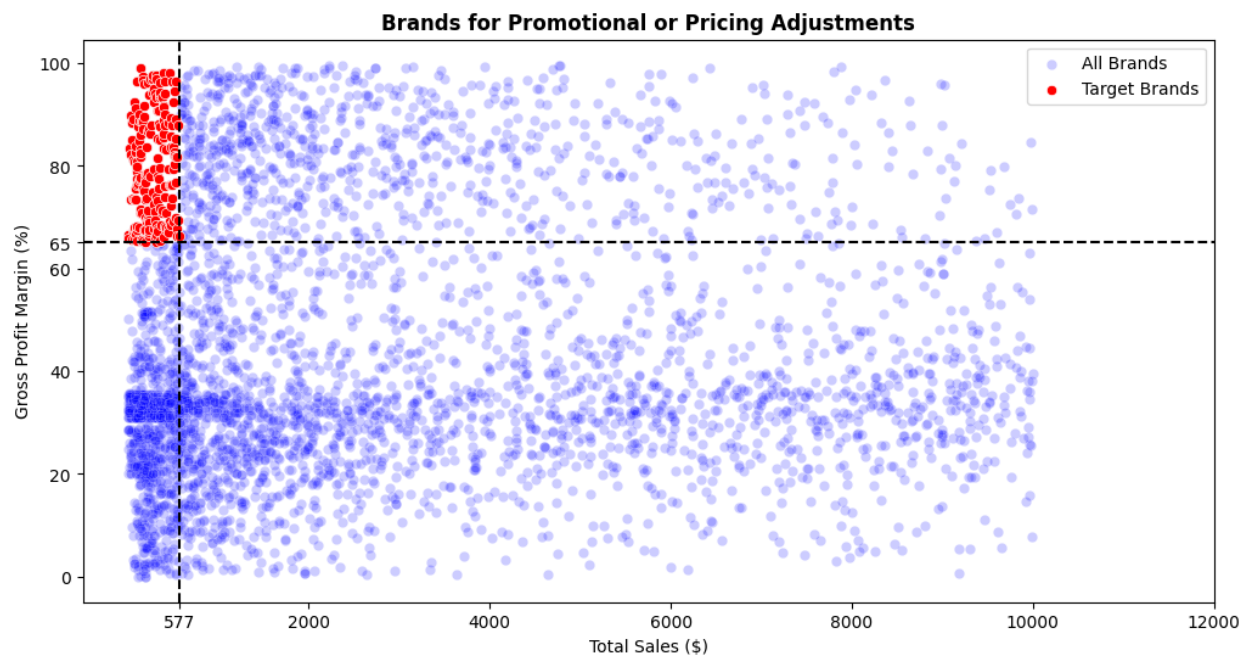
Identifying Business Problem Questions:

- Identify Brands that need Promotional or Pricing Adjustments which exhibit lower sales performance but higher profit margins.
- Which suppliers and brands demonstrate the highest sales performance?
- Which suppliers contribute the most to total purchase dollars?
- Compare the total procurement value of the top 10 suppliers with that of all remaining suppliers.
- Does purchasing in bulk reduce the unit price, and what is the optimal purchase volume for cost savings?
- Which suppliers have low inventory turnover, indicating excess stock and slow-moving products?
- Which suppliers are responsible for the largest share of unsold inventory value?
- Identify the 95% confidence intervals for profit margins of top-performing and low-performing vendors.
- Is there a significant difference in profit margins between top-performing and low-performing suppliers?

Business Problem Solutions:

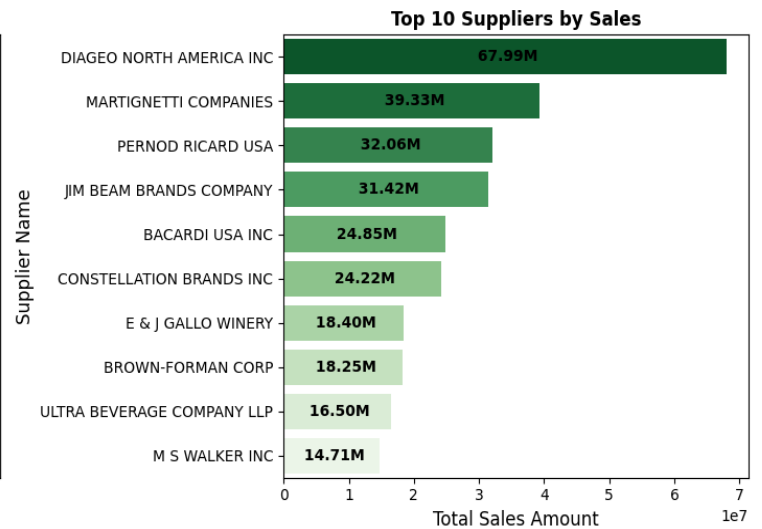
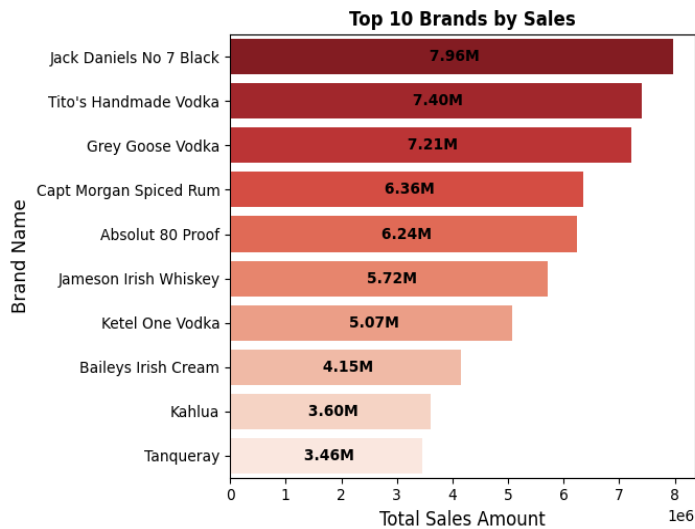
a) Identified Brands that needed Promotional or Pricing Adjustments which exhibit lower sales performance but higher profit margins.

- Figured total sales amount below 15% threshold and profit margin above 85% threshold to identify brands with low sales but high profit margins
- Targeted those brands who had total sales amount less than 577.74 but had gross profit margin higher than 65.04 %
- **Some of the targeted brands - Debauchery Pnt Nr, Crown Royal Apple, Santa Rita Organic Svgn BI**
- Visualized with the help of scatterplot to figure out targeted brands.



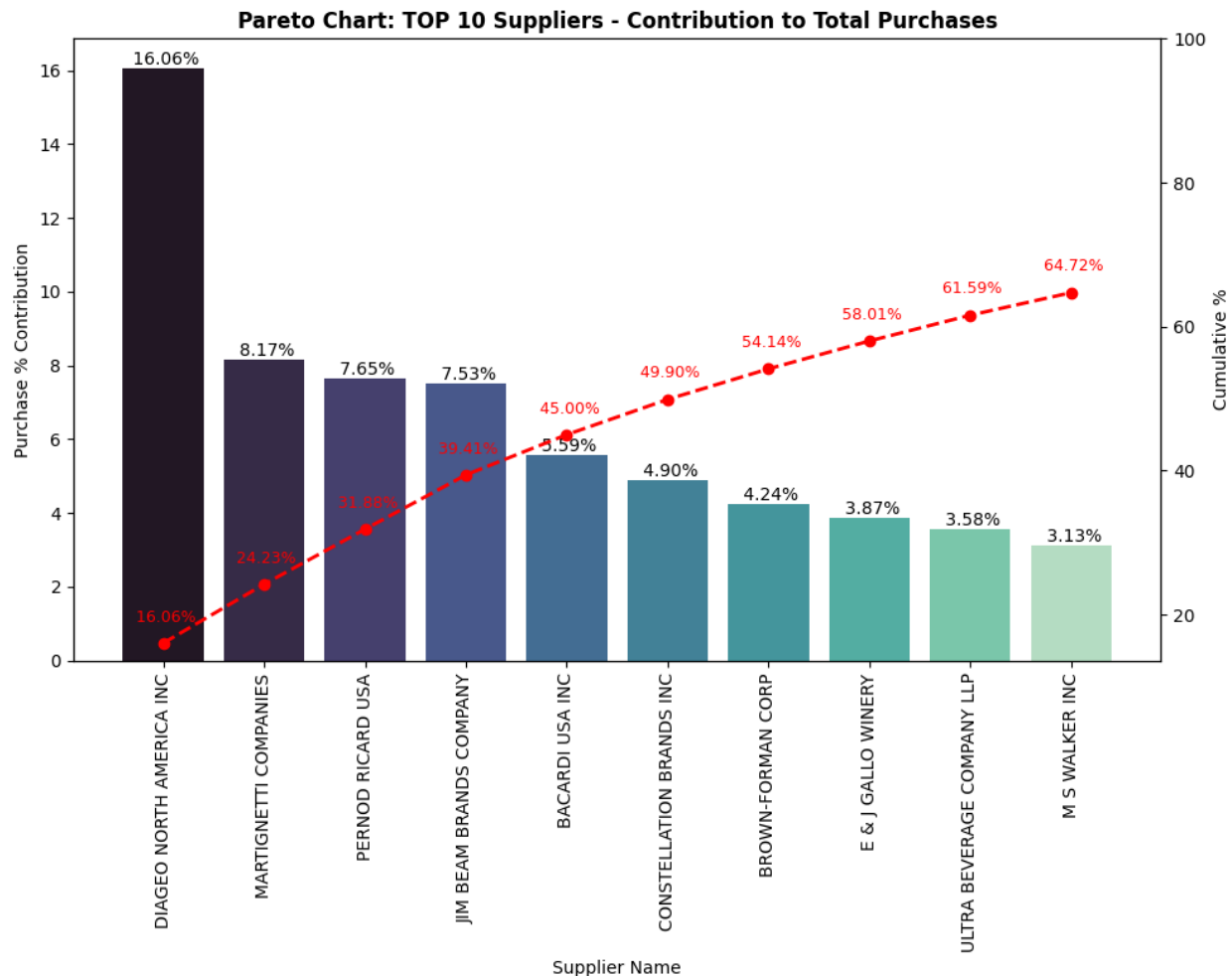
b) Found suppliers and brands who demonstrated the highest sales performance.

- Analyzed the top 10 suppliers and brands with the highest total sales amount.
- **Top suppliers - Diageo North America Inc, Martignetti Companies etc.**
- **Top Brands - Jack Daniels No 7 Black, Tito's Handmade Vodka etc.**
- Visualized with the help of bar plot



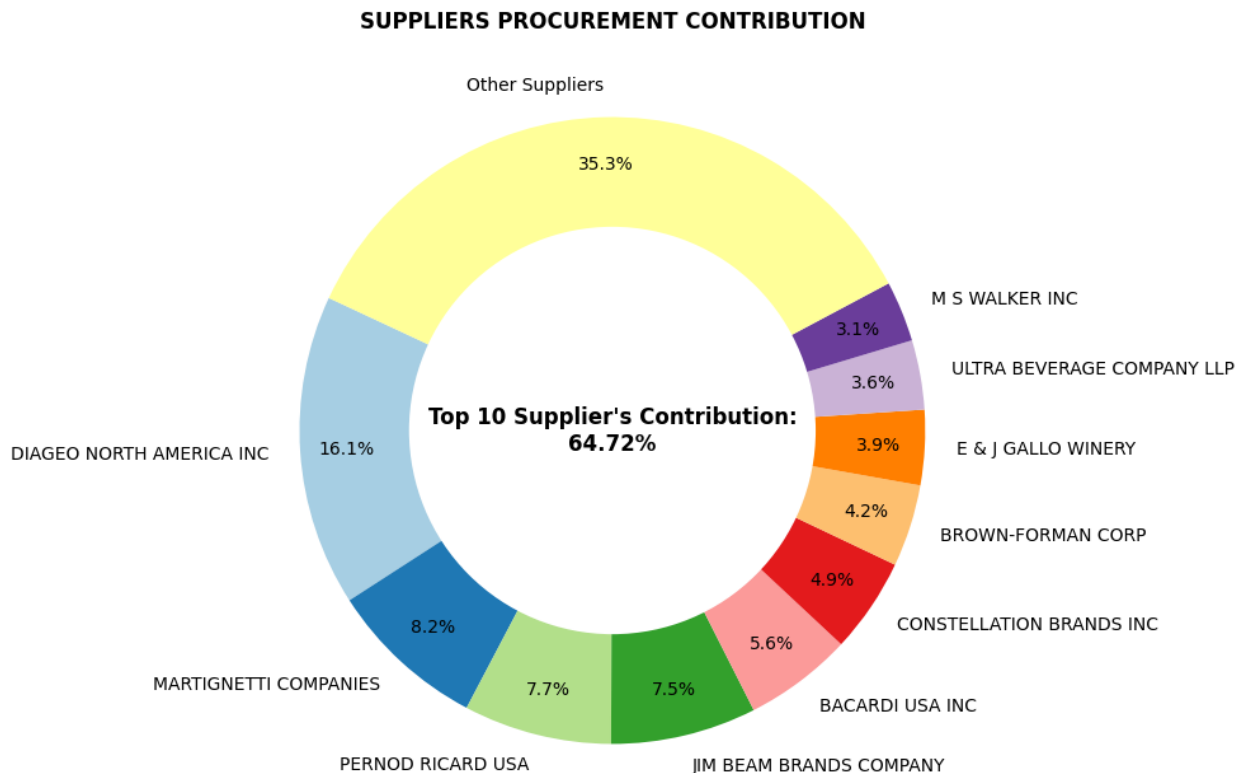
c) Found suppliers who contributed the most to total purchases amount.

- Found the suppliers with the highest purchase percentage contribution.
- Also included the cumulative percentage contribution of the Top 10 suppliers.
- **Top Contributors - Diageo North America Inc, Martignetti Companies etc.**
- **The Top 10 suppliers contributed to around 65% of total purchases made.**
- Visualized with the help if Pareto Chart.



d) Compared the total procurement value of the Top 10 suppliers with that of all remaining suppliers.

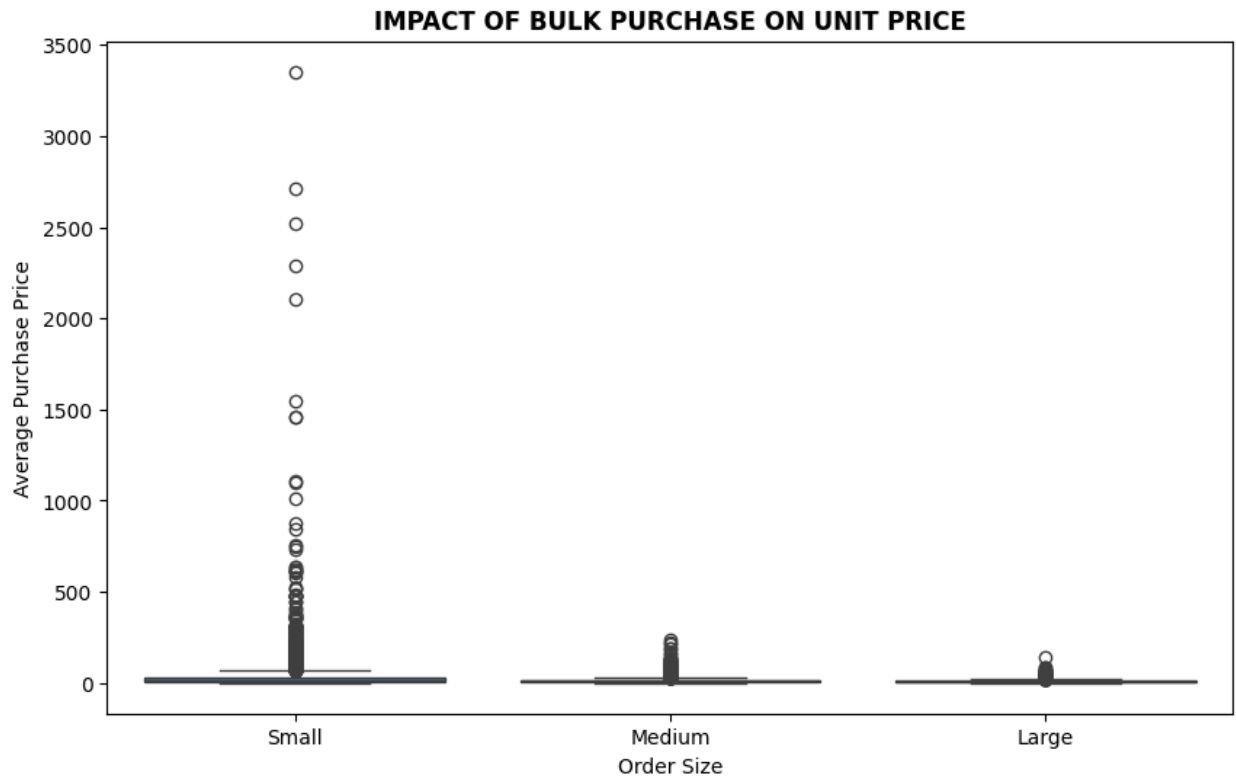
- Found the ratio of total procurement value between Top 10 suppliers and rest of other suppliers.
- **Around 65% of Procurement was made by Top 10 suppliers whereas other suppliers contributed only 35%.**
- Visualized with the help of donut chart.



e) Analyzed whether purchasing in bulk reduces the unit price, and what is the optimal purchase volume for cost savings?

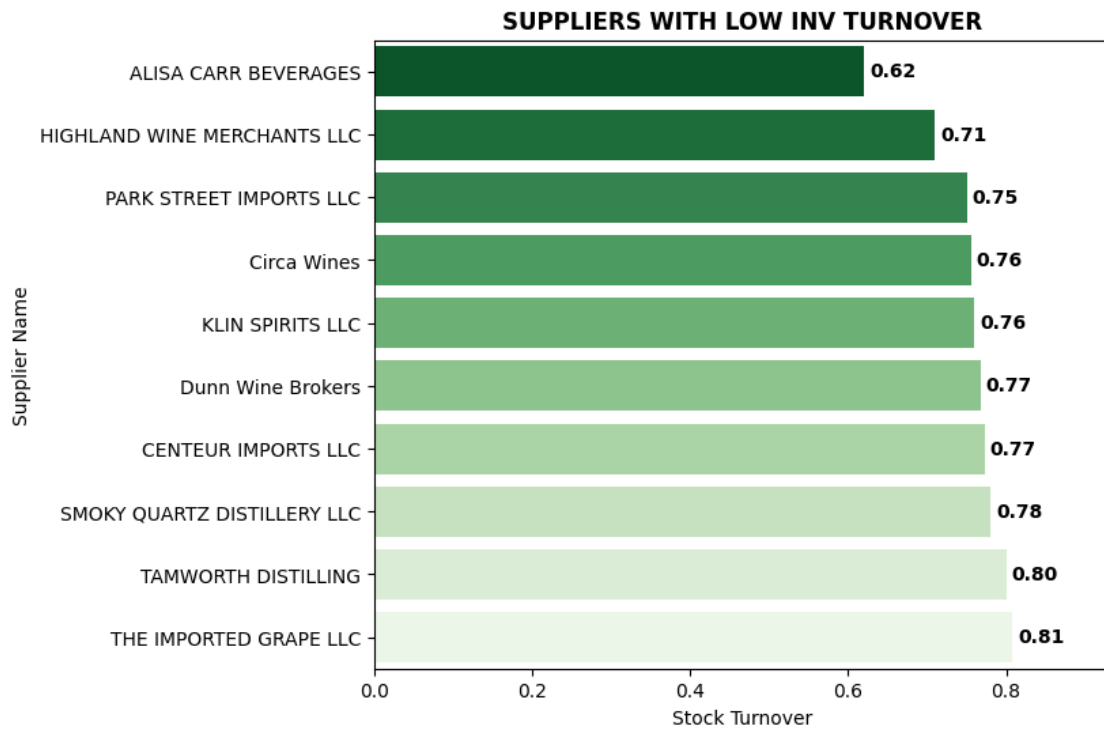
- Distinguished order size as small, medium, large based on total purchases made
- Identified the average purchase price of each order size.
- **Large's Category exhibited the lower average price along with a narrower range of values in the boxplot, indicating bulk purchases reduce the average purchase price per unit, making it optimal purchase volume.**

- The price difference between Small and Large orders is significant, with a **74% reduction in unit cost**.
- Visualized with the help of boxplot to look at range of values (identifying number of outliers).



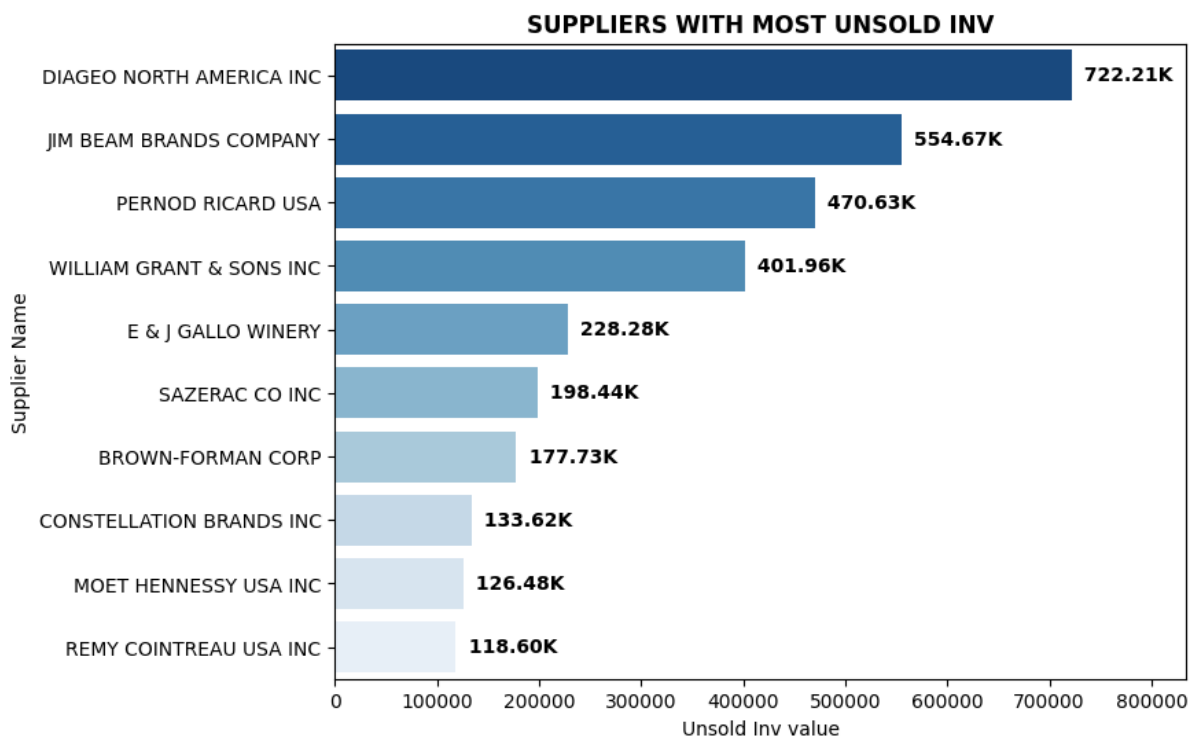
f) Identified suppliers with low inventory turnover, indicating excess stock and slow-moving products

- Filtered out the data with suppliers having turnover < 1 and then found out the Top 10 suppliers with lowest average stock turnover.
- **Alisa Carr Beverages had the lowest inventory turnover of 0.62**
- Visualized with the help of bar plot



g) Identified the suppliers who were responsible for the largest share of unsold inventory value.

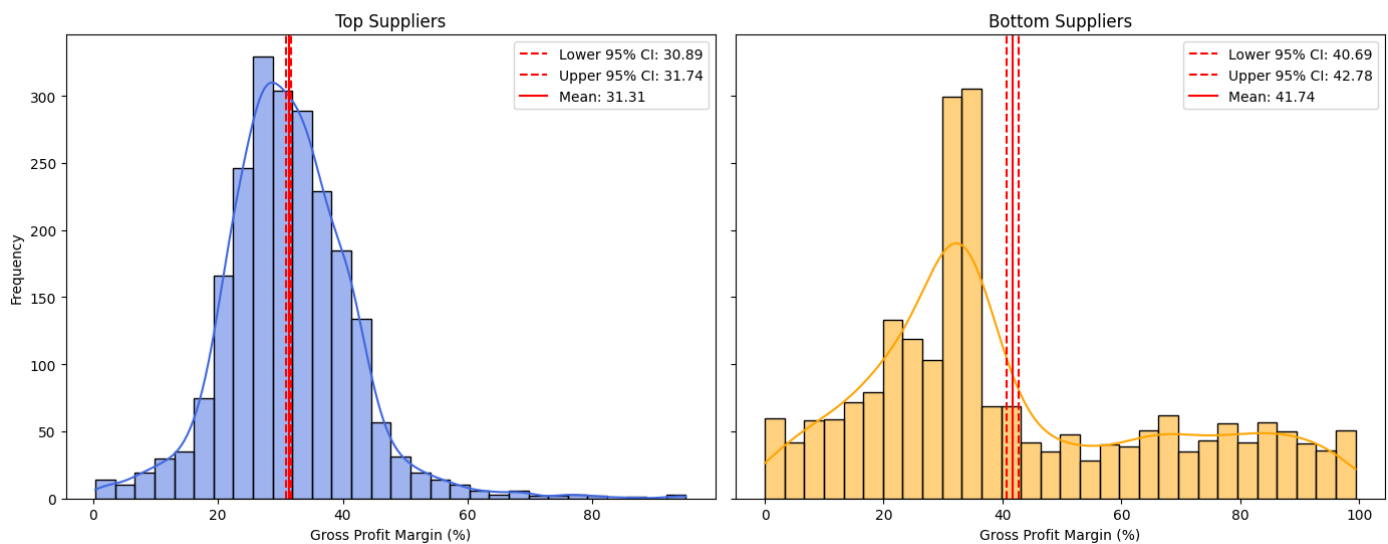
- Calculated Unsold Inventory Value
- Found VinSpire currently has \$2.6 million worth of capital tied up in inventory.
- **Top Contributors - Diageo North America Inc, Jim Beam Brands Company etc.**
- Visualized with the help of bar plot



h) Identified the 95% confidence intervals for profit margins of top-performing and low-performing suppliers.

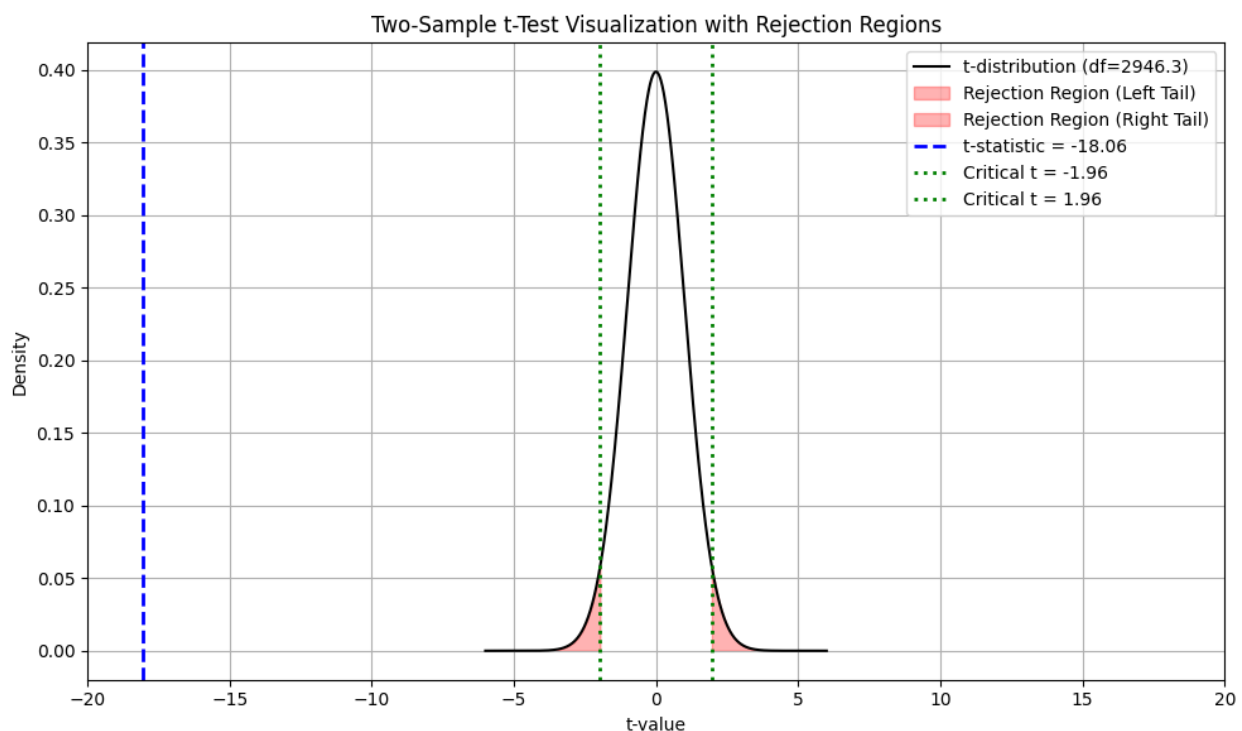
- Figured out high and low threshold for total sales amount
- Suppliers having sales higher than high threshold are top performing whereas suppliers having sales less than low threshold are low performing.
- Filtered out the GPM for high and low performing suppliers
- Calculated Confidence Interval
- Found Mean - 31.31, CI – (30.89, 31.74)
- **The Top suppliers had an average profit of 31.31% and we are 95% sure that their GPM lies between 30.89% and 31.74%**
- **The Bottom suppliers had an average profit of 41.74% and we are 95% sure that their GPM lies between 40.69% and 42.78%**
- **This indicates that low-volume suppliers may be leveraging premium pricing, operating in niche markets, or benefiting from lower operational costs.**
- Visualized with the help of histogram using kernel density estimator.

Gross Profit Margin Distribution



i) Hypothesis Testing - Analyzed whether there was a significant difference in profit margins between top-performing and low-performing suppliers.

- H0 (Null Hypothesis): There is no significant difference in the mean profit margins of top-performing and low-performing suppliers.
- H1 (Alternative Hypothesis): The mean profit margins of top-performing and low-performing suppliers are significantly different.
- Suppliers having sales higher than high threshold are top performing whereas suppliers having sales less than low threshold are low performing.
- Two sample t-test for top and low performing suppliers was conducted.
- **Findings: T-Stats: -18.0578, P-Value: 0.0000**
- The t-statistic of -18.06 shows that top-performing suppliers have lower average profit margins than low-performing ones, with a difference about 18 times greater than random variation.
- The p-value (0.0000) < 0.05, meaning **we reject the null hypothesis** concluding there is significant difference in profit margins.
- **Overall, there is a statistically significant difference in profit margins between top- and low-performing suppliers.**
- Visualized using `plt.plot(x,y)` do show a bell curve.

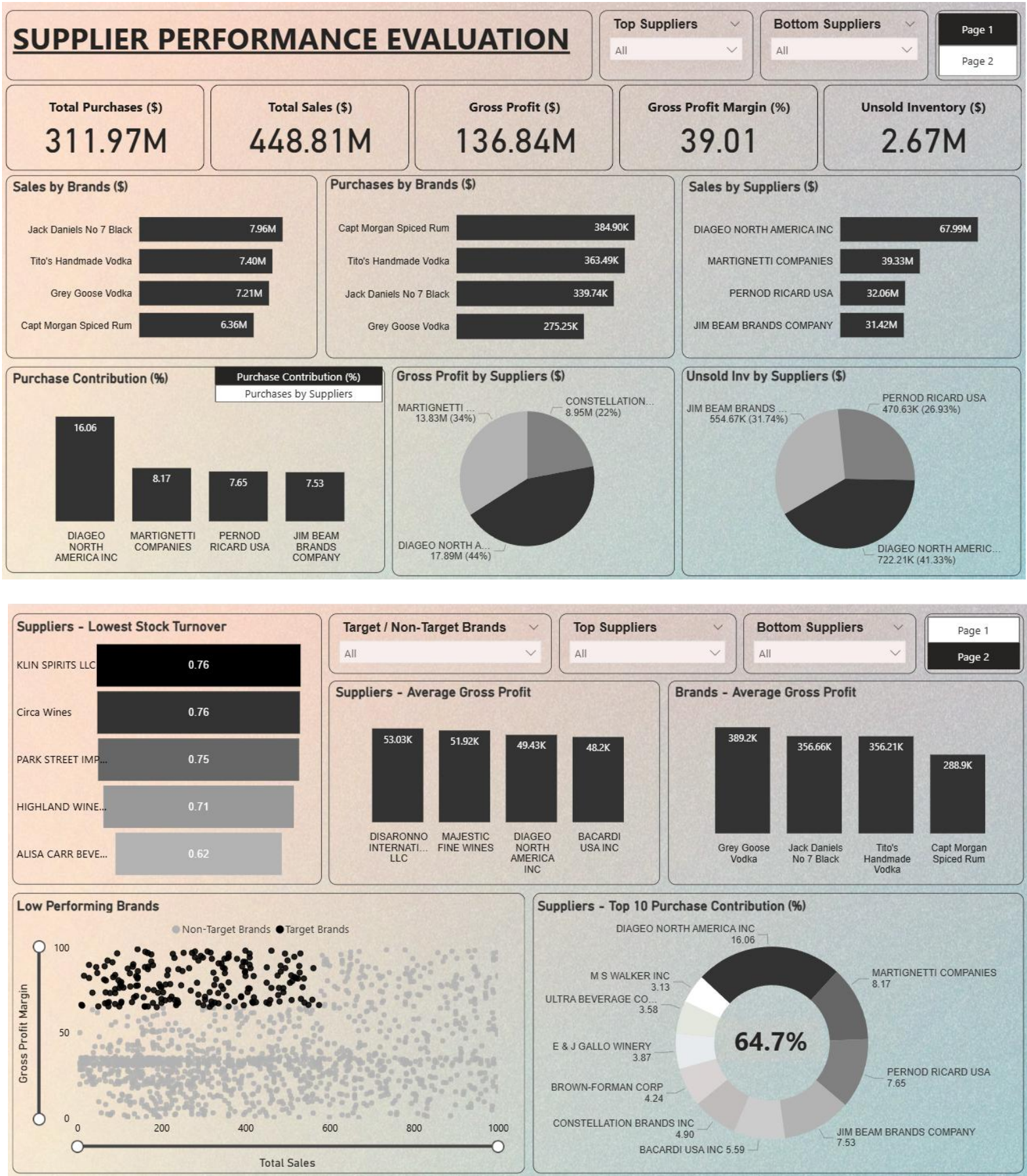


Final Recommendations:

- **Optimize pricing** for low-sales, high-margin brands to increase sales volume without compromising profitability.
- **Diversify supplier partnerships** to reduce dependence on a limited number of suppliers, enhancing supply chain resilience and minimizing risks associated with disruptions.
- **Leverage bulk purchasing** to obtain better pricing discounts, reduce per-unit costs, and optimize inventory levels, thereby improving overall cost efficiency and competitiveness.
- **Reduce slow-moving stock** by analyzing and adjusting purchase order quantities to better match demand, implementing targeted clearance sales to quickly free up inventory space, and improving storage management practices to optimize warehouse efficiency and minimize holding costs.
- **Implement just-in-time (JIT) inventory** practices alongside consolidating the supplier base to reduce excess inventory, improve turnover, and free up capital tied in stock.
- **Collaborate with low-performing suppliers** to enhance their market reach through competitive pricing, targeted marketing initiatives, and improved distribution channels to increase sales volume and drive mutual profitability.
- **Leverage advanced demand forecasting and inventory analytics** to align procurement with market demand and reduce excess stock.

Implementing these recommendations will enable VinSpire to drive sustainable profitability, mitigate risks effectively, and enhance operational efficiency across the organization.

Dashboard Overview:



The above dashboard provides a comprehensive overview of VinSpire's procurement and inventory performance. It highlights key metrics such as supplier contributions, inventory levels, and sales trends. The visualizations emphasize the concentration of procurement among top suppliers, inventory tied up due to multiple suppliers, and the performance of both high-margin and low-sales brands. This dashboard supports data-driven decision-making to optimize pricing strategies, diversify supplier partnerships, leverage bulk purchasing, and improve inventory management—aligned with the recommendations outlined to enhance profitability, reduce risks, and increase operational efficiency.